11 The Reproduction Room

Two ways to make the next generation

CORE PATH · BSC	EXTENSION · MSC	BEFORE THIS ROOM
about 25 min	**+ 28 min**	Nothing beyond the README. The Copying Room (01) explains why copying errors are switched off here.
steps 1–4	steps 5–7	

OPEN THE ROOM

`https://evolinus.github.io/zooe/teaching/evolutionary-laboratories/#reproduction`

Two founders reproduce side by side for four generations. On the left, every individual makes two exact clones; on the right, two parents make two offspring, each with one recombined copy from each parent. This guide asks what each way of reproducing buys, and what it costs.

LEARNING OBJECTIVES

After the core path you should be able to

- Describe how asexual and sexual reproduction make the next generation.
- Predict head counts and numbers of distinct genotypes on each side.
- Identify a crossover and explain how recombination creates new combinations.
- State the two-fold cost of sex and the advantage it buys.

The extension adds

- Calculate how the cost of sex compounds over generations.
- Compare the variation created by crossing over with that from independent assortment (2^n gametes).
- Use the full-sib recursion to explain why this sexual line understates what sex generates.

SUGGESTED ROUTES

- **25-minute practical (BSc):** steps 1–4 and questions 1–3.
- **Lecture demonstration (10 minutes):** steps 1 and 2 with the class predicting the two numbers.
- **MSc:** steps 5–7 and questions 4–6 (about 30 minutes), ideally followed by the Linkage Room's Part 2.

How to work through it. Every step asks you to *predict* before you press anything, and to *explain* afterwards. Write the prediction down and commit to it: a wrong prediction you can account for teaches more than a right one you guessed. Runs are random, so where a step says so, run it more than once before you decide what happened.

1 Numbers

5 min

SET UP
Nothing to set. Press **Reset** if a run is already on screen.

PREDICT FIRST
Both sides start with two individuals. How many individuals will each side have after four generations?

RUN
Press **Auto-play** and watch both panels to the end.

LOOK AT
Population now under each heading.

EXPLAIN
Why does one side double each generation while the other only replaces itself?

2 Variety

8 min

SET UP
The run you just made.

PREDICT FIRST
How many genetically *distinct* individuals will each side have at the end?

RUN
Read *distinct genotypes* under each heading. Press **Reset** and **Auto-play** again.

EXPLAIN
The asexual side has many more individuals. Why does it not have more genotypes? Why does the sexual side keep producing genotypes that have never existed before?

3 Find a crossover

7 min

SET UP
Any finished run.

PREDICT FIRST
In a sexual offspring, what will a chromosome copy look like if a crossover happened during meiosis in its parent?

RUN
Look for a circle on the right where one half changes shade partway down. Step through with **Next generation** if that is easier.

EXPLAIN
Describe in your own words what happened at that point in meiosis, and why the asexual circles never show it.

4 The trade-off

5 min

SET UP

Nothing to change.

PREDICT FIRST

In a stable environment, which strategy will leave more descendants? In a changing one?

EXPLAIN

Use the room's example of aphids and water fleas to explain why many species do both.

EXTENSION · MSC

5 The cost compounds

10 min

SET UP

Nothing to set; this step is a calculation checked against the run.

PREDICT FIRST

Asexual $N(t) = 2^{t+1}$, sexual $N(t) = 2$. Calculate the ratio after 3, 10 and 20 generations.

RUN

Check the first value against the panels at generation 3.

EXPLAIN

The room assumes males contribute nothing but genes. How would paternal care change the size of the cost? Give an example.

6 How much variety does the drawing leave out?

10 min

SET UP

Any run.

PREDICT FIRST

Each individual here has one chromosome pair and one crossover. Independent assortment alone gives 2^n gametes from n chromosome pairs. How many is that for a human ($n = 23$)? For this room?

RUN

Reset and auto-play three times. Does the sexual side ever produce the same genotype twice?

EXPLAIN

Where does the variety on the right-hand side come from in this room, and where does most of it come from in a real genome?

7 Sibling mating

8 min

SET UP

Look at the sexual side from generation 2.

PREDICT FIRST

The only mates available are siblings. Using $H_t = \frac{1}{2}H_{t-1} + \frac{1}{4}H_{t-2}$ and $H_0 = H_1 = 1$, calculate H for generations 2, 3 and 4.

EXPLAIN

What does this inbreeding do to the variety a sexual line can generate over many generations? Why does the room say its sexual panel *understates* what sex does in a real population?

DISCUSSION AND WORKSHEET QUESTIONS

1. BSc Why is the two-fold cost of sex sometimes called the “cost of males”?
2. BSc A disease that kills one genotype arrives at generation 4. What happens to each side?
3. BSc Bdelloid rotifers have reproduced asexually for millions of years. Why is that considered a puzzle?
4. MSc Explain Muller’s ratchet and why it bites hardest in small asexual populations.
5. MSc Summarise the Red Queen hypothesis for the maintenance of sex. What kind of evidence supports it?
6. MSc Why does recombination need genetic variation to be useful? What would sex do in a population where every individual was identical?

CHECKING YOUR REASONING

Read this after you have written your own answers. These are what the model should show on most runs — not the only acceptable explanations.

- **Step 1.** Asexual: 2 → 4 → 8 → 16. Sexual: 2 throughout. Each asexual individual reproduces alone; a sexual pair must share its offspring.
- **Step 2.** Asexual: still only the 2 founding genotypes. Sexual: 2 individuals, but both new genotypes that never existed before, because crossover positions vary continuously.
- **Step 3.** The half that changes shade is one chromosome copy built from pieces of both of its parent's copies; asexual copies are never broken and rejoined.
- **Step 4.** Stable environments favour sheer numbers (asexual); changing ones favour variety (sexual). Facultative switching takes both advantages.
- **Step 5.** Ratios 8, 1,024 and about a million. With paternal care the cost falls, because a male's investment raises the number of offspring that survive.
- **Step 6.** $2^{23} \approx 8.4$ million gametes from assortment alone in a human, and 2 here. Here almost all variety comes from where the crossover falls; in a real genome it comes from both.
- **Step 7.** H = 0.75, 0.625, 0.5. Heterozygosity falls by about a fifth each generation in a full-sib line, so the variety recombination can work with drains away.